

## 4.3.1 Keys and Integrity Constraints

کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

CS XI

Chapter 4: Data and Analysis

Lecture 04 of 15

★ Golden

### 4.3.1 Keys and Integrity Constraints ★ Golden

#### LEARN IT

A **key** is a field (or a set of fields) that uniquely identifies records and links tables. Integrity constraints are the rules that keep those keys honest.

**Candidate key.** Any field (or combination) that *could* uniquely identify a row. A table may have several. In an Employee table: EmpID, EmpLicense# and EmpPassport are all candidate keys.

**Primary key (PK).** The one candidate you actually choose. Rules: unique, never NULL, only one PK per table (it may be a **composite** of two columns). Example: EmpID.

**Alternate key.** The leftover candidate keys. After EmpID is chosen, EmpLicense# and EmpPassport become alternate keys — still unique, just not “the” identifier.

**Foreign key (FK).** A field in this table that refers to the PK of another table. It *may* contain duplicates (many employees in one department) and, in some designs, NULL.

#### WORKED EXAMPLE

**Classroom analogy.** Roll Number, B-Form number and phone number can all identify a student (candidate keys). The school picks Roll Number as the primary key. B-Form and phone become alternate keys. In a **Result** table, RollNo is a foreign key pointing back to Student.

| Table   | Field           | Role                             |
|---------|-----------------|----------------------------------|
| Student | RollNo          | Primary key                      |
| Student | BFormNo         | Alternate key                    |
| Result  | RollNo          | Foreign key → Student.RollNo     |
| Result  | ExamID + RollNo | Could be a composite primary key |

#### EXAM TIP

Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the DBMS could not tell rows apart.

#### COMMON MISTAKE

A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL). Roll numbers are issued by the school and stay unique.

#### اردو میں سمجھیں

کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔ پرائمری کی منتخب شدہ شناخت ہے — منفرد اور خالی نہیں۔ جو بچ جائیں وہ آلٹرنیٹ کیز ہیں۔ فارن کی دوسرے ٹیبل کی پرائمری کی طرف اشارہ کرتی ہے اور ٹیبلز کو جوڑتی ہے۔

#### CHECK YOURSELF

1. Can a table have two primary keys?
2. Can a foreign key have duplicate values?

Answers: 1. No — only one PK (it may be composite). 2. Yes — many child rows may point to the same parent.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** A primary key:

- a. Uniquely identifies each record
- b. May freely repeat
- c. Is always a name
- d. Is a chart

**MCQ 2.** A candidate key is:

- a. Any field set that could be the PK
- b. A foreign key
- c. A report
- d. A form

**MCQ 3.** An alternate key is:

- a. A candidate not chosen as PK
- b. A duplicate PK
- c. A null PK
- d. A pie

**MCQ 4.** A foreign key:

- a. Refers to a PK in another table
- b. Is always unique in its table
- c. Sorts the table
- d. Encrypts the DB

**MCQ 5.** NULL in a primary key is:

- a. Not allowed
- b. Required
- c. A default
- d. A relationship type

**MCQ 6.** Integrity constraints exist to:

- a. Stop invalid data
- b. Draw graphs
- c. Route packets
- d. Compile Python

## Extra short questions

1. Define PK, FK, candidate, alternate.
2. Give a two-table example with an FK.
3. Entity integrity rule.
4. Why not use student name as PK?

## Extra long question

1. For Members and Borrow in a library, choose keys and write two integrity rules. Show one invalid insert the DBMS should reject. ★ Golden.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

### **Short**

1. PK: chosen unique identifier. Candidate: any unique identifier. Alternate: unused candidate. FK: copy of another table's PK.
2. Students(roll PK); Marks(roll FK, subject, score).
3. PK unique and not NULL.
4. Names duplicate and change.

### **Long**

1. MemberID PK; Borrow(BorrowID PK, MemberID FK). Reject Borrow with unknown MemberID or NULL MemberID. Mention cascade later.

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Next lecture: 4.3 Relational Database Management System (RDBMS)